VAE

Auto-Encoding Variational Bayes

Kingma & Welling | arXiv:1312.6114v11

Turn intractable latent-variable learning into end-to-end stochastic optimization by combining amortized variational inference with a differentiable sampling path.

variational inference · latent variables · reparameterization

1 Problem

- Directed models with continuous latent variables often have an intractable marginal likelihood and posterior.
- Per-example MCMC or iterative variational updates are too expensive for large datasets and minibatch learning.

2 Probabilistic autoencoder

ENCODE A DISTRIBUTION

- The recognition model q_phi(z|x) amortizes posterior inference; the decoder p_theta(x|z) defines the generative likelihood.
- A standard Gaussian prior organizes a continuous latent code that supports sampling, denoising and visualization.

3 Learning objective

MAXIMIZE THE ELBO

- `ELBO = E_q[log p_theta(x|z)] - KL(q_phi(z|x) || p(z))`
- The reconstruction term explains data; the KL term regularizes each approximate posterior toward the prior.

4 Key trick

MOVE RANDOMNESS OUTSIDE

- `z = mu_phi(x) + sigma_phi(x) * epsilon; epsilon ~ N(0,I)`
- Reparameterization produces a low-variance differentiable Monte Carlo estimator optimized with ordinary stochastic gradients.

5 Evidence

MNIST + FREY FACES

- AEVB converges faster and reaches a better variational bound than wake-sleep across tested latent dimensions.
- Experiments learn smooth 2-D manifolds, generate samples and scale with minibatches using about one sample per datapoint.

6 Meaning & limits

- The paper connects probabilistic inference and neural autoencoders, establishing the modern VAE recipe.
- The original estimator targets reparameterizable continuous latents; simple Gaussian decoders can yield blurry samples and loose bounds.